

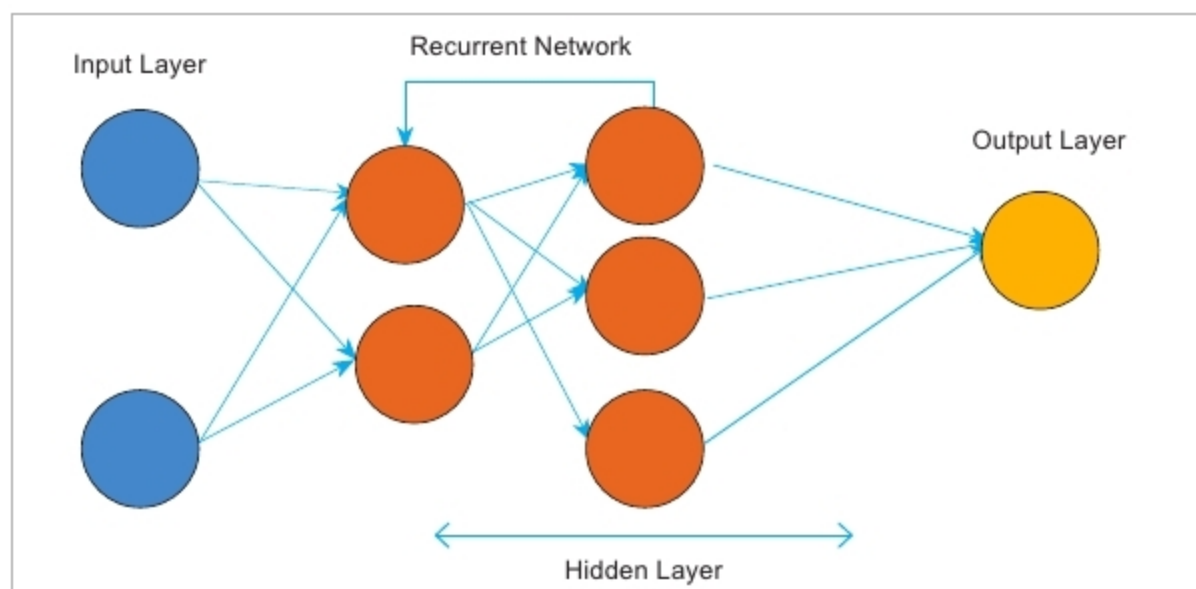


Figure 2: Recurrent neural network architecture

reduction, classification, regression, collaborative filtering, feature learning and topic modelling.

Deep learning architectures and data sets

The various deep learning architectures are based on discriminative models and generative models. Discriminative models are widely used by the industry, while the generative models are still evolving. Let's examine the key features of these models.

Discriminative models: Here, we explore the architectures of few discriminative models like CNN and RNN with respect to computer vision and speech, or text processing.

- **Convolutional neural network (CNN) based models:** The architecture is designed based on the number of layers the model has, as well as a high-level description of the layers and techniques used in the network. A typical diagram of the CNN based model will look like what's shown in Figure 1.

This figure depicts the two-layer neural network — one hidden layer of four neurons and one output layer with two neurons and three inputs. AlexNet, VGG and ResNet are variations in terms of the network layer stacking and the new network elements of the base CNN network architecture.

- **Recurrent neural network (RNN) based models:** These are designed

to act upon the series of data with variable length. They remember the previous input (using the in-memory state) to be used along with the input for the next node. They are best suited for applications related to speech and text processing, and are based on natural language understanding (NLU). The variation of RNN is the Long Short Term Memory (LSTM) network.

Generative models: Generative adversarial networks (GAN) are generative deep neural network models intended to produce or generate new content from given training.

Data sets: The data sets are used for training the neural network. Generally, the training needs large data sets, which are used for image processing, natural language processing (NLP) and speech processing. The most popular data sets are COCO, ImageNet and Open Images.

The applicability of deep learning architectures to industrial domains

The following are some of the industrial domains to which deep learning architectures are applicable.

- **Healthcare:** Deep learning and its innovations are advancing the future of precision medicine and health management. Breast cancer and skin cancer diagnosis are just a few examples of deep learning in healthcare. In the years to come,

computer aided diagnosis will play a major role in this domain.

- **Computer vision and pattern recognition:** Image detection, segmentation, multi-object detection, and tracing on videos and on live stream, are a few examples of using deep learning.
- **Computer games, robots:** Advances in deep reinforcement learning are creating the possibilities of introducing AI into games (like Go) and robots.
- **Automated driving:** This includes computer vision, reinforcement learning and many other subjects used in the control of autonomous vehicles.
- **Voice-activated intelligent assistants:** Apple's Siri, Google Now and Microsoft's Cortana are a few examples of deep learning in voice search and voice-activated intelligent assistants.
- **Advertising:** Deep learning enables publishers and ad networks to leverage their content to create data-driven predictive advertising, precisely targeted advertising, and much more.
- **Predicting natural calamities:** Predicting natural hazards and setting up a deep learning based emergency alert system will play an important role in the coming years.
- **Finance:** Analysing trading strategies, reviewing commercial loans and form contracts, and preventing cyber-attacks are examples of deep learning in the finance industry.
- **Aerospace and defence:** Deep learning is used to detect/locate as well as mask objects from other satellites that locate areas of interest, and identify safe or unsafe zones for troops.

Open source frameworks for deep learning

Here is a brief introduction to a few open source frameworks for deep learning.

TensorFlow: This is an end-to-end library, low level and high-end framework for deep learning. It can handle deep neural networks for image